Yuda Risma Wahyudi

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Education

National Taipei University of Technology — Taipei, Taiwan
Master of Science in Automation Technology. GPA 4.00 — September 2019 – August 2021

State University of Malang — Malang, Indonesia
Bachelor of Engineering in Electrical Engineering. GPA 3.70 — July 2014 – December 2018

Experience

Senior Automation Test Engineer — September 2022 - present
PEGATRON — Batam, Indonesia

- Designed and developed automated test programs using C# for production line FATP (Final Assembly Test and Pack).
- Conducted software testing on Linux-based systems and software applications for efficient product quality assurance.
- Developed and maintained reusable testing libraries and frameworks, improving code maintainability and efficiency.
- Proficient in utilizing Git for version control and collaboration.
- Improved and contributed to the testing device driver codebase in C# and Python. The driver includes RF power meters, signal analyzers, signal generators, and DAQ devices, through various communication protocols (Modbus, TCP/IP, SSH, UART, etc.)
- Contributed to successful NPI projects, particularly on 5G related products, by conducting thorough RF testing for WIFI 5, WIFI 6, 5G (FR1 and FR2), and LTE technologies.
- Responsible for the design and optimization of the NR OTA station, ensuring accurate and efficient testing of wireless devices.
- Proficient in utilizing various RF test instruments such as Litepoint IQxel, RSA, and Keysight power meter.

Research Assistant — August 2021 – August 2022
Assistive Robotic Laboratory, National Taipei University of Technology — Taipei Taiwan

- Led and collaborated on industry-funded research projects involving automated industrial systems.
- Conducting research in deep learning and image processing. Developed expertise in object detection, object classification, and 3D pose detection using relevant methods.
- Designed and implemented a human-robot collaboration system leveraging Media Pipe and Open Pose for accurate human-skeleton detection.
- Established robot communication protocols. Developed robot API to enable seamless communication between computers and various industrial robot platforms (FANUC, YASKAWA, Denso, and Staubli)
- Leveraged C++ and Python, including object-oriented programming principles, to effectively develop and implement complex robotic control systems, optimization algorithms, and machine learning methods. This expertise contributed to enhanced robot performance and functionality in various research projects.
- Contributed to the publication of 3 papers and 2 US patents related to the optical inspection and robotic control systems.

Head of Robotic Team — July 2016 – June 2018
Robotic Laboratory, State University of Malang — Malang, Indonesia

- Led and managed a cross-functional team (mechanical, electrical, and programming) for the design and development of a hexapod robot system.
- Directed the development of the robot's core functionalities. Designed and implemented the control system and kinematic model using advanced algorithms, ensuring efficient and precise robot motion.
- Electrical system design and schematic, assigning tasks and guiding team members for PCB design and electrical wiring.
- Utilized expertise in 3D modeling software (SOLIDWORKS) to create the robot's mechanical design, fostering effective collaboration with the mechanical team.
- Experienced in working with various microcontrollers (MCUs) for embedded systems, including STM32, AVR, and Raspberry Pi. Possess expertise in integrating these MCUs with actuators and sensors to achieve desired functionalities.

Skills

Programming	: C++, C#, Python, VB.Net, ROS
Development Tools	: Visual Studio, VS Code, PyCharm, Git and Github
Engineering Software	: MATLAB, SOLIDWORK
Languages	: English (proficient), Indonesian (native), Chinese (conversational)